



Technical Service Bulletin: tsb090070

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Diesel Exhaust Fluid (DEF) Exhaust Blockage (Crystallization/Polymerization) and Fault Codes 1682, 2973, 2772, 2773, 3548, 3538, 3568, 3582, and 5655

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Core Issue

Product Affected

- All midrange engines equipped with aftertreatment selective catalyst reduction (SCR)

Symptoms:

- Exhaust pipe diameters reduced by diesel exhaust fluid (DEF) crystallization or polymerization and/or DEF crystals visible externally.
- Fault Codes 1682, 2973, 2772, 2773, 3548, 3538, 3568, 3582, and 5655.

Root Cause:

- DEF crystals can build up in exhaust system for a number of reasons, including:
 - Pinholes in welds, loose clamps, and/or leaks in OEM exhaust piping.
 - Injecting DEF with low air flow to nozzle supply line
 - Engines with low exhaust temperature duty cycles, such as a transit bus or yard spotter
- DEF crystals occur first and then turn into a polymer with prolonged temperature exposure below 290°C [554°F].

Confirmation

Crystallization or polymerization can lead to four types of exhaust piping blockage/restriction.

1. Less than 20% DEF Accumulation (See Figure 1 below)
 - Minor tracking marks can be seen in the decomposition tube.
 - This is normal and will burn off in normal operating conditions during an active regeneration.
 - Vehicle performance **not** impacted.
 - No further action required.
2. 20 to 40 % DEF accumulation (See Figure 2 below)
 - This is normal and will burn off in normal operating conditions during an active regeneration.

- Vehicle performance **not** impacted.
 - No further action required.
3. 40 to 50 % DEF Accumulation (See Figure 3 below)
- Soft Crystal Buildup - White in color.
 - Will dissolve easily using warm water.
 - Covering 30 to 50 percent of exhaust cross section.
 - Can form when diesel exhaust fluid evaporates. (Example poor atomization of DEF).
 - Can be made more severe by exhaust system design violation.
 - Vehicle will clear in service during active regeneration.
 - Perform a Stationary regen.
 - Check aftertreatment system gaskets and clamps for leaks and replace as required.
 - Vehicle performance **not** impacted.
 - No further action required.
4. 50 to 80 % DEF Accumulation (See Figure 4 below)
- Polymerization occurs when crystals are heated for prolonged periods of time at temperatures below 290°C [554°F].
 - White in color.
 - **Not** able to be dissolved using warm water.
 - Decomp tube will need to be removed and cleaned out.
 - Check aftertreatment system gaskets and clamps for leaks and replace as required.
 - Duty cycle will need to be assessed and potential stationery regeneration schedule implemented.
 - Hard and difficult to remove.
 - Covering 50 to 80 percent of exhaust cross section.
 - Usually forms at nozzle location, at point where injection spray touches the exhaust wall.
 - Can be made more severe by exhaust system design violation.

Additional Items

- External Buildup
 - Indication an exhaust malfunction exists that will also cause internal DEF buildup
- Internal Decomposition Pipe Cleaning when crystallization leak is witnessed externally (DEF Injector Gasket). (See Figure 6 below).
 - External Crystal growth can be witnessed from the DEF Injector gasket.
 - There is no detriment to the performance of the system when this is witnessed, no additional repairs are required other than to change the Injector Gasket.
 - New gasket has been released to remedy these leaks.
 - Cummins Part Number 5508734.
 - CES Part Number A063J608.
- DEF injector changes, Injectors seen to be “wet” or with a “drip” during system prime.
 - During the prime procedure a build-up of fluid will be noted from the DEF injector nozzle – as seen in Figure 7.
 - This is a normal condition and does **not** point to a potential injector issue.
 - The injector tip is a “wet tip”, Small amounts of fluid are released to allow any crystal build up to be dissolved. During the priming of the system.
- Duty Cycle and Exhaust System Design Considerations

- DEF accumulations are more likely in low temperature duty cycles below 290°C [554°F].
- Euro 4 and Euro 5 exhaust system designs where location of aftertreatment DEF dosing nozzle is less than 100 mm [3.9 in] downstream and/or less than 270 mm [10.6 in] upstream of a bend in piping and/or less than 440 mm [17.3 in] from inlet end of exhaust gas processor are more prone to DEF accumulation.
- **Only** when duty cycle or exhaust system design are the cause of a separate customer complaint can Cummins Inc. class complaint as crystallization or polymerization.

Resolution

- DEF accumulation covering less than 50% of cross sectional area does **not** need to be removed.
- DEF accumulation above 50% **must** be removed. See corresponding Service Manual. Reference Procedure 011-062 in Section 11.
- For any instances of reoccurring DEF accumulation where a resolution is **not** found using above information, a Technical Service Request (TSR) is to be raised.

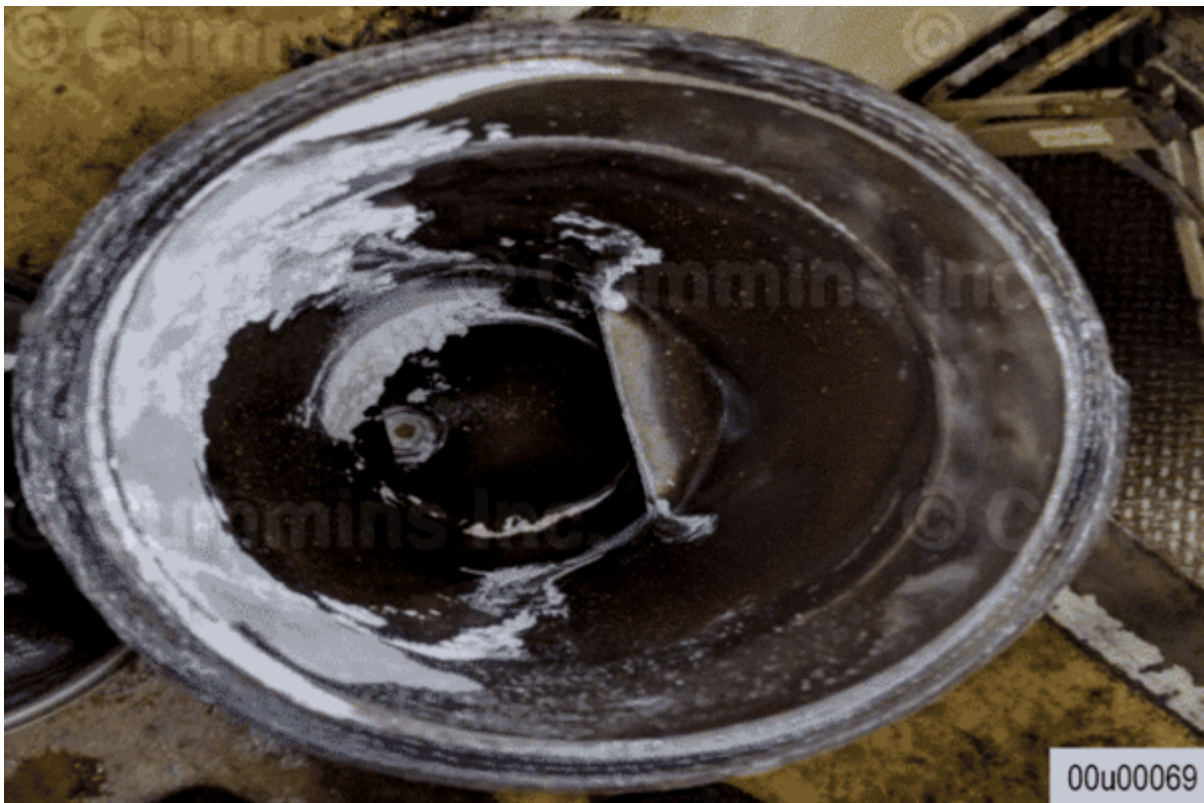


Figure 1, Less than 20% DEF Accumulation.



Figure 2, 20% to 40% DEF Accumulation.

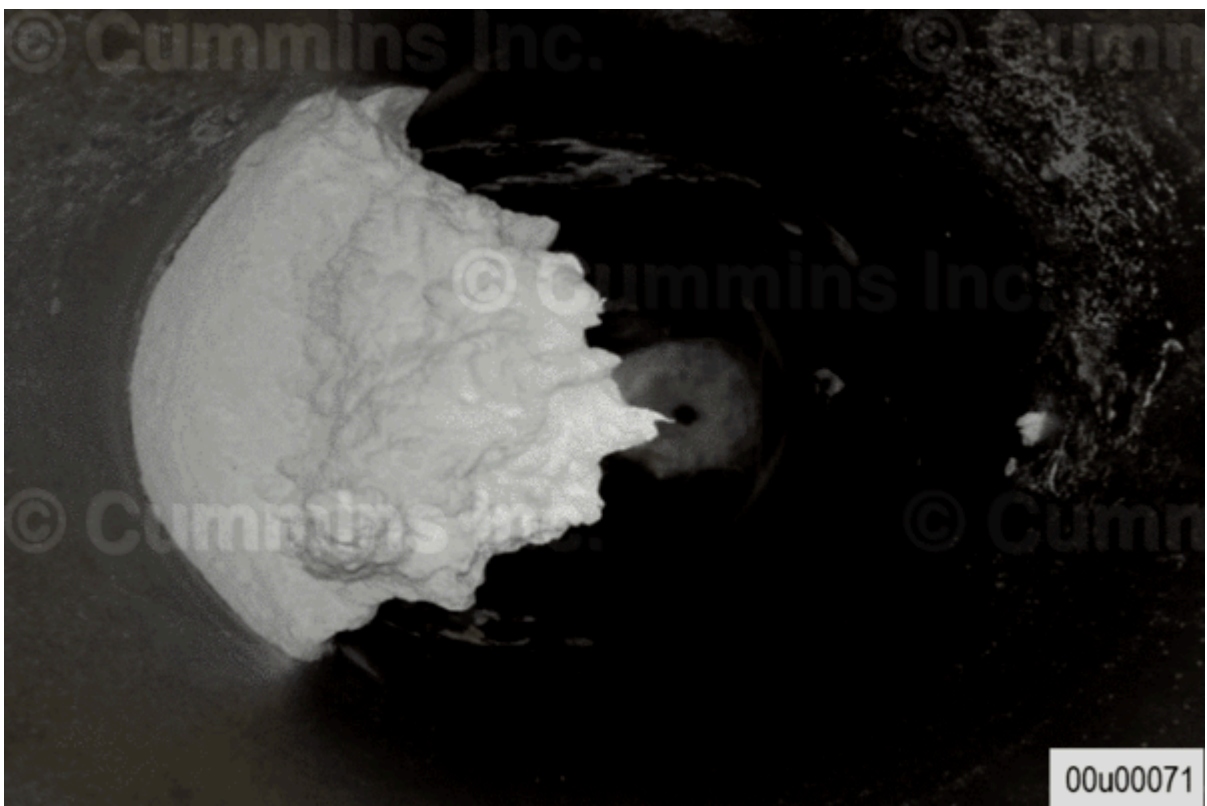


Figure 3, 40% to 50% DEF Accumulation.



Figure 4, 50% to 80% DEF Accumulation.



Figure 5, External Buildup.



Figure 6, Buildup of Fluid Noted from the DEF Injector Nozzle.

Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Document History

Date	Details
2017-5-23	Non-Product Problem Solving (PPS)
2020-1-30	Made necessary edits to TSB.
2020-1-31	Added new graphics.
2020-2-13	Made necessary edits to TSB.
2021-10-14	Added mention of yard spotter.
2021-11-4	Adde Fault Code 5655.

Last Modified: 12-Nov-2021
